

Correlation, Regression & Curve Fitting

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Introduction to Correlation

➤ What is Correlation?

Correlation is a statistical measure that shows the **degree and direction of the relationship** between two variables.

➤ Key Idea

When one variable changes, the other also changes.

Correlation does **not** mean cause and effect — it only shows **association**.

➤ Examples

Height and Weight

Temperature and Electricity Consumption

Study Hours and Marks

➤ Purpose

To understand how two variables move together.

To study strength of the relationship (strong, weak, none).

To help in prediction and analysis.

Types of Correlation

➤ Positive Correlation

Both variables move in the **same direction**.

(If X increases → Y increases)

➤ Negative Correlation

Variables move in **opposite directions**.

(If X increases → Y decreases)

➤ Zero Correlation

No relationship exists between the variables.

➤ Linear Correlation

Data follows a straight-line pattern.

➤ Non-Linear Correlation

Relationship is curved (not a straight line).

➤ Simple Correlation

Relationship between **two variables only**.

➤ Multiple Correlation

Relationship of **one variable with two or more variables**.

Scatter Diagram

➤ What is a Scatter Diagram?

A scatter diagram is a **graphical method** of studying correlation by plotting paired data points on a graph.

➤ Key Points

- Each point represents a pair (X, Y).
- The pattern of points indicates the type of correlation.
- Easy visual method to judge relationship strength.

➤ Interpretation

- **Upward pattern** → Positive correlation
- **Downward pattern** → Negative correlation
- **No clear pattern** → Zero correlation

Definition

Karl Pearson's correlation coefficient (r) measures the **strength and direction** of linear correlation between two variables.

➤ Formula

$$r = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sqrt{[\sum(x - \bar{x})^2 \sum(y - \bar{y})^2]}}$$

➤ Features

- Most widely used method
- Applicable for quantitative data
- Measures only **linear** correlation

Key Properties

Range: $-1 \leq r \leq +1$

- **$r = +1$** → Perfect positive correlation
- **$r = -1$** → Perfect negative correlation
- **$r = 0$** → No linear correlation

Other Points

- **r is independent of origin and scale**
(Adding or multiplying constants doesn't affect r)
- **r shows degree of linear relationship**
- **r does not imply causation — only association**

Definition

Spearman's rank correlation measures the **degree of association** between two ranked variables.

Formula

$$r = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

When to Use

- When data is in **ranks**
- When exact values are unavailable
- When Pearson's method cannot be applied

Spearman's Rank Method (When Ties Occur)

What are Ties?

When two or more values are equal, they get the **same rank**.

Handling Ties

- Assign **average ranks**
- Use corrected formula (if needed)

Uses

- Useful in competitions, ratings, preferences, surveys.

Regression (Introduction)

What is Regression?

Regression studies the **dependence of one variable on another** to make predictions.

Purpose

- To estimate unknown values
- To predict outcomes
- To analyze cause and effect (approximate)

Regression Lines

Types of Regression Lines

- **Regression of Y on X**
Predicts Y for given X
- **Regression of X on Y**
Predicts X for given Y

- **General Forms**

Y on X: $y - \bar{y} = b_{yx}(x - \bar{x})$

X on Y: $x - \bar{x} = b_{xy}(y - \bar{y})$

Regression Coefficients

Formulas

$$b_{yx} = r (\sigma_y / \sigma_x)$$

$$b_{xy} = r (\sigma_x / \sigma_y)$$

Meaning

- Shows **rate of change**
- Helps in constructing regression lines

Properties of Regression Coefficients

Important Properties

$$b_{yx} b_{xy} = r^2$$

- One coefficient $> 1 \Rightarrow$ the other must be < 1
- Independent of **origin**
- Not independent of **scale**

Difference: Correlation vs Regression

Correlation

- Measures strength of relationship
- No prediction
- r shows direction (+ or -)

Regression

- Predicts unknown values
- Gives regression equations
- Shows functional dependence

Curve Fitting (Introduction)

Definition

- Curve fitting finds a **mathematical curve** that best fits the given data.

Need

- To understand trend
- To forecast values
- To smooth raw data

Least Squares Method

Concept

The best-fit curve is obtained by **minimizing** the sum of squared errors:

$$\Sigma (\text{Observed} - \text{Estimated})^2 \rightarrow \text{Minimum}$$

Why Use It?

- Simple
- Reliable
- Provides most accurate curve for given data

Straight Line Fitting (Linear Fit) Equation

$$y = a + bx$$

Normal Equations

$$\Sigma y = na + b\Sigma x$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2$$

Use

- Data showing linear pattern
- Basic trend analysis

Quadratic Curve Fitting

Equation

$$y = a + bx + cx^2$$

Use When

- Data shows a **curved pattern**

Idea

Fit a **parabolic curve** using least squares.

Exponential Curve Fitting Equation

$$y = Ce^{(ax)}$$

Method

- Take log on both sides
- Convert to linear form
- Apply least squares method

Used For

Growth and decay patterns.

Applications of Regression

Used For

- Predictions and forecasts
- Trend analysis
- Estimating unknown values
- Machine learning basics

Examples

- Predicting sales
- Estimating future demand

Summary of Unit–1

In This Unit, We Learned:

- Types and importance of correlation
 - Pearson & Spearman correlation
 - Meaning of regression and regression lines
 - Regression coefficients and properties
 - Curve fitting using least squares
 - Linear, quadratic, and exponential models
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- ✓ Foundation for statistics & data analysis
 - ✓ Useful for real-world prediction and modeling

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